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Project Category:	☐ Research oriented	☐ Application oriented	Batch ID:	AIML-B1
	☐ Product oriented	☐ Other (please specify)		
Specify if others:				

Title of the Project: Design and Development of a Trustworthy Agentic Foundation Model Framework for Explainable Legal Reasoning and Decision Intelligence Using Multi-Agent Retrieval-Augmented Generation and Dynamic Legal Knowledge Graphs

Abstract

This research addresses the challenges of developing reliable, explainable, and trustworthy Artificial Intelligence systems for complex legal reasoning and decision support. Existing Legal AI systems face critical limitations, including hallucinated legal interpretations, limited reasoning transparency, inadequate citation verification, static knowledge repositories, poor adaptability to evolving legislation, and insufficient mechanisms for trust assessment and explainability. These limitations restrict the effective adoption of AI-assisted legal systems in high-stakes legal environments where accuracy, accountability, transparency, and evidence-based reasoning are essential. To address these challenges, this research proposes an integrated framework that combines Foundation Models, Multi-Agent Systems, Retrieval-Augmented Generation, Dynamic Legal Knowledge Graphs, Explainable AI, and Trustworthy AI principles for reliable legal reasoning and decision intelligence

This research proposes a Trustworthy Agentic Foundation Model Framework for Explainable Legal Reasoning and Decision Intelligence using Multi-Agent Retrieval-Augmented Generation and Dynamic Legal Knowledge Graphs. The proposed framework introduces a modular agentic architecture in which specialized intelligent agents collaboratively perform legal document retrieval, legal knowledge graph exploration, evidence verification, legal reasoning, explainability generation, and trust evaluation under the coordination of a central orchestration agent. Unlike existing Legal AI systems that primarily focus on retrieval or response generation, the proposed framework integrates Dynamic Legal Knowledge Graphs with Foundation Models to enable structured legal reasoning over continuously evolving statutes, judicial precedents, constitutional provisions, and regulatory documents. A Retrieval-Augmented Generation pipeline will be employed to minimize hallucinations by grounding responses in authenticated legal sources, while an explainability module will generate evidence-supported reasoning pathways to improve transparency and user confidence.

To enhance the reliability of AI-assisted legal decision-making, the framework incorporates a comprehensive Trust Evaluation Module that estimates citation validity, legal consistency, evidence coverage, confidence scores, and hallucination risk. Human-in-the-loop feedback mechanisms will further facilitate continuous learning and system refinement through expert validation. The proposed framework will be implemented using modern AI technologies, including Foundation Models, Multi-Agent orchestration frameworks, vector databases, graph databases, and Retrieval-Augmented Generation pipelines. Its performance will be evaluated using publicly available legal benchmark datasets through comprehensive experiments involving legal question answering, case retrieval, legal summarization, legal reasoning, and decision support tasks. Evaluation will consider both conventional performance metrics (Accuracy, Precision, Recall, F1-score, BLEU, ROUGE, BERTScore)

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and trust-oriented metrics such as citation accuracy, explanation quality, hallucination rate, evidence consistency, and user trust.

The expected contributions of this research include the development of a novel Trustworthy Agentic Legal AI architecture, an adaptive Dynamic Legal Knowledge Graph model, a multi-agent legal reasoning framework, explainable legal decision intelligence mechanisms, and a comprehensive trust evaluation methodology. The proposed research is expected to advance the state of the art in Legal Artificial Intelligence by integrating trustworthy AI, explainable AI, multi-agent systems, and foundation models into a unified framework capable of delivering reliable, transparent, and scalable legal decision support. The outcomes of this research have the potential to contribute to intelligent judicial assistance, legal analytics, policy decision support, legal education, e-governance, and digital justice systems while supporting the development of next-generation trustworthy Legal AI platforms suitable for real-world deployment.

Keywords: Legal Artificial Intelligence, Foundation Models, Multi-Agent Systems, Retrieval-Augmented Generation, Dynamic Legal Knowledge Graphs, Explainable AI, Trustworthy AI, Legal Reasoning, Decision Intelligence, Large Language Models, Legal Question Answering, Digital Justice.

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